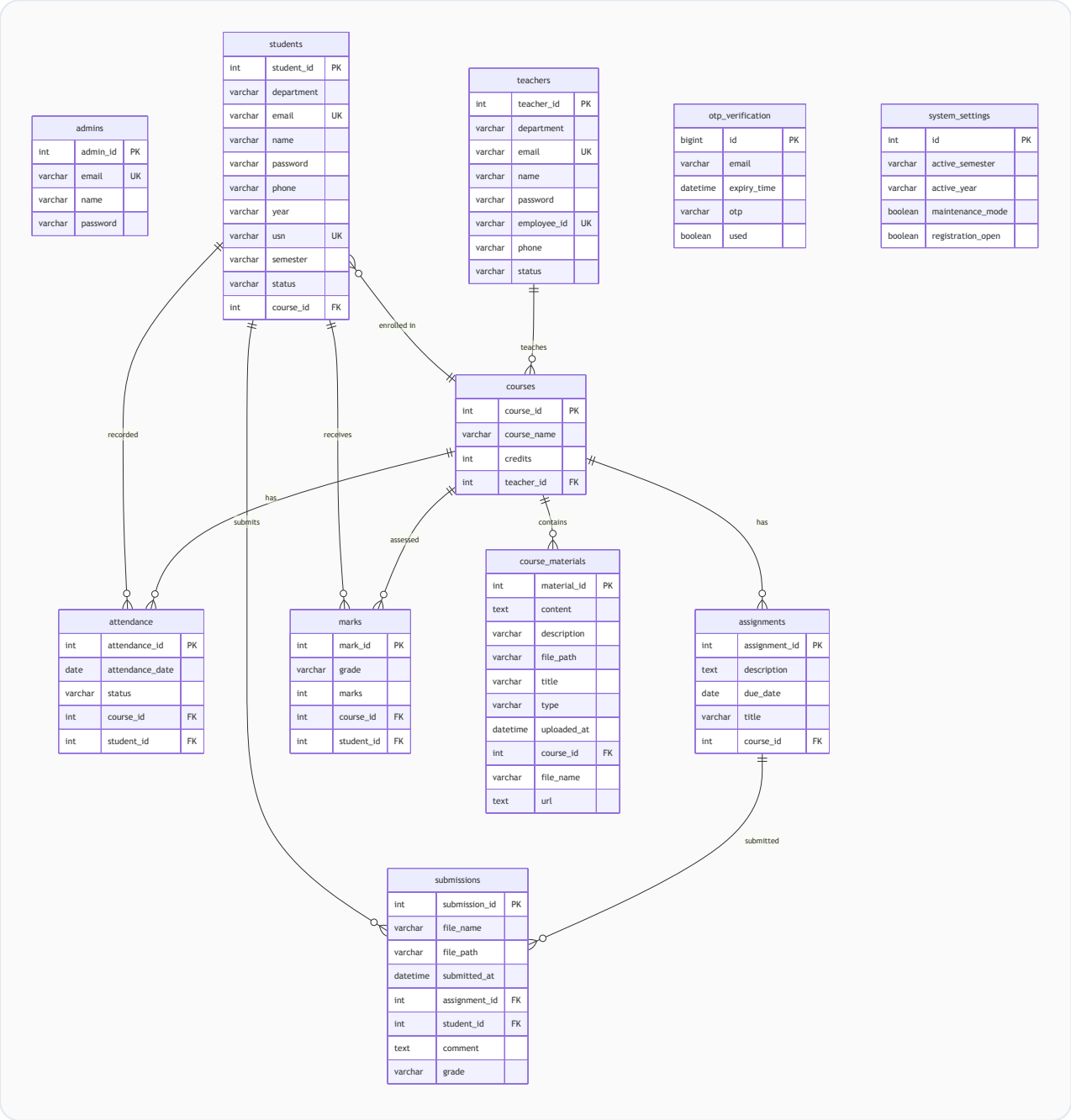
🗄️ Database Schema Document

EduTrack Academic Management Platform VTU Internship 2026 — Team 15 April 2026

ER Diagram



Overview

The EduTrack database consists of **11 tables** organised around three core user types — Students, Teachers, and Admins — with supporting tables for academic data management.

TABLE	DESCRIPTION
admins	Administrator accounts
teachers	Teacher accounts and profiles
students	Student accounts and profiles
courses	Courses created by teachers
attendance	Attendance records per student per course
assignments	Assignments created for courses
submissions	Student assignment submissions
marks	Student marks and grades per course
course_materials	Study materials uploaded by teachers
otp_verification	OTP records for password reset
system_settings	Platform-wide configuration

Table Definitions

1. admins

Administrator account information. Auto-seeded on first startup.

COLUMN	DATA TYPE	CONSTRAINTS	DESCRIPTION
admin_id	INT	PK	Unique admin identifier
email	VARCHAR(255)	UNIQUE, NOT NULL	Admin email
name	VARCHAR(255)	NOT NULL	Admin full name
password	VARCHAR(255)	NOT NULL	BCrypt hashed password

2. teachers

COLUMN	DATA TYPE	CONSTRAINTS	DESCRIPTION
teacher_id	INT	PK	Unique teacher identifier
department	VARCHAR(255)	NOT NULL	Department name
email	VARCHAR(255)	UNIQUE, NOT NULL	Teacher email
name	VARCHAR(255)	NOT NULL	Teacher full name
password	VARCHAR(255)	NOT NULL	BCrypt hashed password
employee_id	VARCHAR(255)	UNIQUE	Employee ID
phone	VARCHAR(255)		Contact number
status	VARCHAR(255)		Active / Suspended

3. students

COLUMN	DATA TYPE	CONSTRAINTS	DESCRIPTION
student_id	INT	PK	Unique student identifier
department	VARCHAR(255)	NOT NULL	Department name
email	VARCHAR(255)	UNIQUE, NOT NULL	Student email
name	VARCHAR(255)	NOT NULL	Student full name
password	VARCHAR(255)	NOT NULL	BCrypt hashed password
phone	VARCHAR(255)		Contact number
year	VARCHAR(255)		Academic year (e.g. 3rd Year)
usn	VARCHAR(255)	UNIQUE	University Seat Number
semester	VARCHAR(255)		Current semester
status	VARCHAR(255)		Active / Suspended / Graduated
course_id	INT	FK → courses	Enrolled course

4. courses

COLUMN	DATA TYPE	CONSTRAINTS	DESCRIPTION
course_id	INT	PK	Unique course identifier
course_name	VARCHAR(255)	NOT NULL	Course name
credits	INT		Credit hours
teacher_id	INT	FK → teachers	Assigned teacher

5. attendance

COLUMN	DATA TYPE	CONSTRAINTS	DESCRIPTION
attendance_id	INT	PK	Unique attendance record
attendance_date	DATE	NOT NULL	Date of attendance
status	VARCHAR(255)		Present / Absent
course_id	INT	FK → courses	Course reference
student_id	INT	FK → students	Student reference

6. assignments

COLUMN	DATA TYPE	CONSTRAINTS	DESCRIPTION
assignment_id	INT	PK	Unique assignment identifier
description	TEXT		Assignment details
due_date	DATE	NOT NULL	Submission deadline
title	VARCHAR(255)	NOT NULL	Assignment title
course_id	INT	FK → courses	Course reference

7. submissions

COLUMN	DATA TYPE	CONSTRAINTS	DESCRIPTION
submission_id	INT	PK	Unique submission identifier
file_name	VARCHAR(255)		Uploaded file name
file_path	VARCHAR(255)		Uploaded file path

COLUMN	DATA TYPE	CONSTRAINTS	DESCRIPTION
submitted_at	DATETIME(6)	NOT NULL	Submission timestamp
assignment_id	INT	FK → assignments	Assignment reference
student_id	INT	FK → students	Student reference
comment	TEXT		Student or teacher comment
grade	VARCHAR(255)		Grade for submission

8. marks

COLUMN	DATA TYPE	CONSTRAINTS	DESCRIPTION
mark_id	INT	PK	Unique mark record
grade	VARCHAR(255)		Auto-calculated grade
marks	INT		Marks scored
course_id	INT	FK → courses	Course reference
student_id	INT	FK → students	Student reference

9. course_materials

COLUMN	DATA TYPE	CONSTRAINTS	DESCRIPTION
material_id	INT	PK	Unique material identifier
content	TEXT		Material content
description	VARCHAR(255)		Short description
file_path	VARCHAR(255)		Storage path
title	VARCHAR(255)	NOT NULL	Material title
type	VARCHAR(255)		Type (file/link)
uploaded_at	DATETIME(6)	NOT NULL	Upload timestamp
course_id	INT	FK → courses	Course reference
file_name	VARCHAR(255)		Original file name

COLUMN	DATA TYPE	CONSTRAINTS	DESCRIPTION
url	TEXT		External link (if any)

10. otp_verification

COLUMN	DATA TYPE	CONSTRAINTS	DESCRIPTION
id	BIGINT	PK	Unique OTP record
email	VARCHAR(255)	NOT NULL	Email OTP sent to
expiry_time	DATETIME(6)	NOT NULL	Expiry (10 min)
otp	VARCHAR(255)	NOT NULL	6-digit OTP code
used	BIT(1)	DEFAULT FALSE	Whether OTP was used

11. system_settings

COLUMN	DATA TYPE	CONSTRAINTS	DESCRIPTION
id	INT	PK	Fixed value: 1
active_semester	VARCHAR(255)		Current semester
active_year	VARCHAR(255)		Current academic year
maintenance_mode	BIT(1)		Blocks logins when true
registration_open	BIT(1)		Allows/blocks student sign-up